

Partially Ordered Sets: A relation R on a set S is called a partial ordering if it is reflexive, antisymmetric and transitive. That is

- 1) aRa for all $a \in S$ (reflexivity)
- 2) aRb and $bRa \Rightarrow a = b$ (antisymmetry) and
- 3) aRb and $bRc \Rightarrow aRc$ (transitivity) for $a, b, c \in S$.

A set S together with a partial order relation R is called a partially ordered set or a poset. It is denoted by (S, R) .

The relation R is often ~~termed~~ denoted by the symbol $\leq$. A partial order denoted by $x \leq y$, means x precedes y .
 $x < y$, means x strictly precedes y .

Ex: Show that the relation $\geq$ is a partial ordering on the set of integers, $\mathbb{Z}$.

- Since (1) $a \geq a$ for every a , $\geq$ is reflexive
 (2) $a \geq b$ and $b \geq a \Rightarrow a = b$, $\geq$ is antisymmetric.
 (3) $a \geq b$ and $b \geq c \Rightarrow a \geq c$, $\geq$ is transitive.

It follows that $\geq$ is a partial ordering on the set of integers and $(\mathbb{Z}, \geq)$ is a poset.

Ex: Consider $P(S)$ as the power set i.e., the set of all subsets⁽²⁾ of a given set S . Show that the inclusion relation $\subseteq$ is a partial ordering on the power set $P(S)$.

→ Since (1) $A \subseteq A$ for all $A \subseteq S$, $\subseteq$ is reflexive.
(2) $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$, $\subseteq$ is antisymmetric.
(3) $A \subseteq B$ and $B \subseteq C \Rightarrow A \subseteq C$, $\subseteq$ is transitive.

It follows that $\subseteq$ is a partial ordering on $P(S)$ and $(P(S), \subseteq)$ is a poset.

Ex: Show that the set $\mathbb{Z}^+$ of all true integers under divisibility relation forms a poset.

→ Since (1) n/n for all $n \in \mathbb{Z}^+$, $/$ is reflexive.
(2) n/m and $m/n \Rightarrow n=m$, $/$ is antisymmetric.
(3) n/m and $m/p \Rightarrow n/p$, $/$ is transitive.

It follows that $/$ is a partial ordering on $\mathbb{Z}^+$ and $(\mathbb{Z}^+, /)$ is a poset.

Note: On the set of all integers, the above relation is not a partial order as a and $(-a)$ both divide each other but $a \neq -a$, i.e., the relation is not antisymmetric.

Comparability: The elements a and b of a poset $(S, \preceq)$ are called comparable if either $a \preceq b$ or $b \preceq a$.

When a and b are elements of S such that neither $a \preceq b$ nor $b \preceq a$ i.e., a and b are not related, a and b

are called incomparable. For ex., in the poset $(\mathbb{Z}^+, /)$ the

integers 2 and 4 are comparable, since $2/4$ but 3 and 5 are incomparable because neither 3 divides 5 nor 5 divides 3.

Def (S, $\leq$) is a poset and every two elements of S are comparable, S is called a totally ordered or linearly ordered set, and $\leq$ is called a total order or a linear order.

A totally ordered set is also called a chain. For ex, in the poset $(\mathbb{Z}, \leq)$ $a \leq b$ for all integers a and b, hence $(\mathbb{Z}, \leq)$ is totally ordered but the poset $(\mathbb{Z}^+, |)$ is not totally ordered since it contains elements that are incomparable such as 3 and 5.

Although an ordered set S may not be totally ordered, it is still possible for a subset A of S to be totally ordered. Clearly, every subset of a totally ordered set S must also be totally ordered. Consider the poset $(\mathbb{Z}^+, |)$ which is not totally ordered but $A = \{2, 6, 12, 36\}$ is a totally ordered subset of $\mathbb{Z}^+$ since $2|6$, $6|12$ and $12|36$.

Note: A chain is said to be anti-chain if no two distinct elements of the set are related.

Product and Lexicographic order:

Product order: For $s, s' \in S$ and $t, t' \in T$, the product set $S \times T$ defined by $(s, t) \leq (s', t')$ if $s \leq_1 s'$ and $t \leq_2 t'$ in T, where (s, t) and $(s', t') \in S \times T$ is called the product order.

[P.T.O.]

order like, life, limitation, limit.

First order the strings by length as
like, life, limit, limitation

Here life $\prec$ like since $l=l$, $i=i$ and $f \prec k$ i.e., f precedes k in English letters.

Next life $\prec$ like $\prec$ limit since $l=l$, $i=i$ and $k \prec m$

The string limit is of length 5 and limitation is of length 10 and $\min(5, 10) = 5$. The first 5 positions are same, hence

life $\prec$ like $\prec$ limit $\prec$ limitation.

Immediate successor: Let $(X, \prec)$ be a poset and suppose $x, y \in X$, then y is said to be immediate successor of x if $x \prec y$ and $x \preceq z \preceq y \Rightarrow x=z$ or $z=y$.

Note: i) We also say y covers x or x is an immediate predecessor of y .

ii) There is no intermediate element between x and y which is distinct from both.

iii) Immediate successor or predecessor may or may not exist for any given elements.

~~Representation~~

Hasse diagrams: A partial order $\preceq$ on a set X can be represented by means of a diagram known as Hasse diagram of $(X, \preceq)$.

We represent the elements of X by points and if y is an immediate successor of x , we take y at a higher level than x and join x and y by a straight line. A diagram formed as above is known as Hasse diagram. Thus there will not be any horizontal lines in the diagram of a poset.

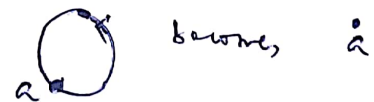
Note: The Hasse diagram for a total order relation can be drawn as a single vertical chain. (16)

Constructing a Hasse Diagram:

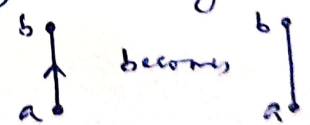
To obtain a Hasse diagram from a directed graph of a partial order relation:

We know that a partial order relation is reflexive and hence its directed graph has loop on each vertex in the diagram. Consequently these loops can be dropped from the directed graph of a partial order relation since they must be present. Because a partial ordering is transitive, the edges showing the implied transitivity can be removed. If we assume that all the edges are pointed upwards then we do not have to show the direction of the edges. In general we can represent a partial ordering on a finite set using the following procedure.

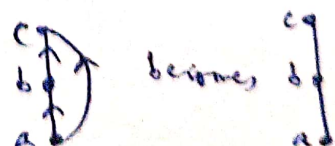
- 1) Start with a directed graph of the relation.
- 2) Remove the loops at all the vertices i.e.,



- 3) If aRb , then b appears above the element a and the element a connected to element b by an edge with arrows pointing upwards. Remove all the arrows.



- 4) Remove all edges whose existence is implied by the transitive property. aRb and $bRc \Rightarrow aRc$.



To recover the directed graph of a partial order relation from the Hasse diagram, the following procedure can be used.

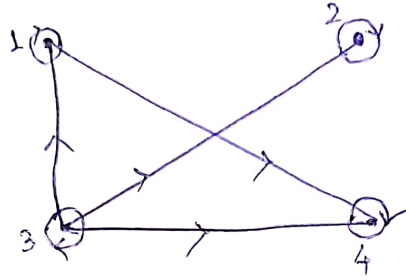
- 1) Reinsert the direction markers on the arrows making all arrows upward.

A set with an ordering relation is well-order if every non-empty subset of the set has a least element.

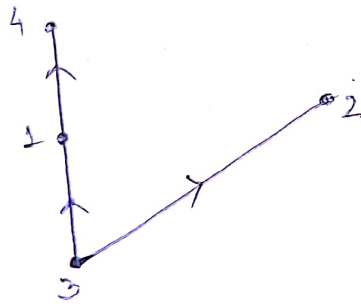
2) Add loop at each vertex.

3) For each sequence of pointing edges from one point to a second point and from second point to third point, add an edge from first to the third pointing towards third point. (9)

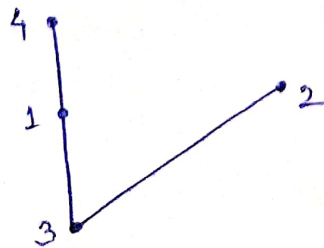
Ex: Draw a Hasse diagram from the directed graph G for a partial order relation on a set $A = \{1, 2, 3, 4\}$.



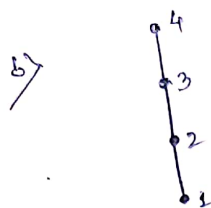
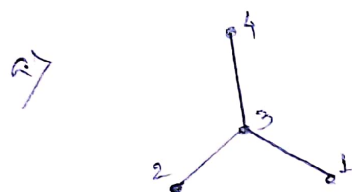
First we remove all the loops at the vertices 1, 2, 3 and 4 and the edge between the vertices 3 and 4 which is transitively implied. Arranging all the arrows pointing up we get the diagram as



Now, removing all the arrows we get the required Hasse diagram as shown below



Ex: Describe the order pairs in the relation determined by the Hasse diagram of a poset $(A, \leq)$ on the set $A = \{1, 2, 3, 4\}$

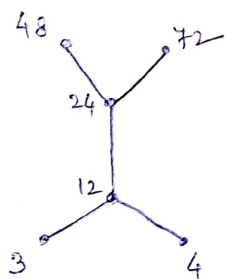


→ Since the relation on A is a partial order, all reflexive pairs $(1,1), (2,2), (3,3), (4,4)$ must be in $\leq$. All edges when converted to upwards towards their vertices give the ordered pairs $(1,3), (2,3), (3,4)$. The transitivity implied arcs give the ordered pairs $(1,4), (2,4)$. Thus the ordered pairs in the relation represented by Hasse diagram in (a) above is

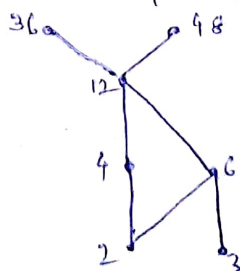
$$\{(1,1), (2,2), (3,3), (4,4), (1,3), (2,3), (3,4), (1,4), (2,4)\}$$

Similarly, the ordered pairs in the relation represented by Hasse diagram (b) is $\{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$.

Ex: Draw the Hasse diagram of $(A, \leq)$ where $A = \{3, 4, 12, 24, 48, 72\}$ and relation $\leq$ be such that $a \leq b$ if a divides b .



Ex: Draw the Hasse diagram of the relation $\leq$ defined as "divides" on set B where $B = \{2, 3, 4, 6, 12, 36, 48\}$

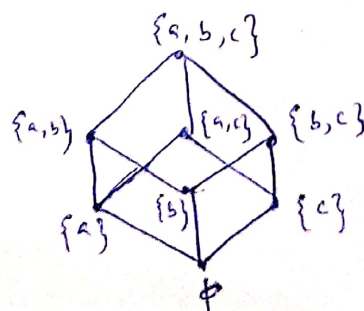


(14)

10a

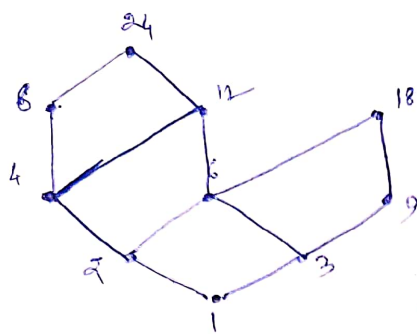


→ Let $S = \{a, b, c\}$, then

$$P(S) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$


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Ex: Let $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ be ordered by the relation "a divides b". Draw the Hasse diagram.

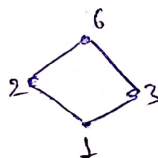


Ex: Draw the Hasse diagram of
 i) $(D_8, |)$ ii) $(D_6, |)$ iii) $A = \{2, 3, 5, 30, 60, 120, 180, 360, '1'\}$
 iv) $E = \{1, 2, 3, 4, 6, 9, '1'\}$

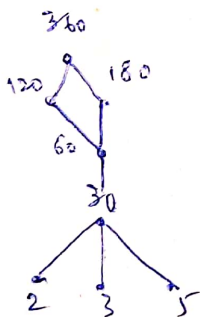
i) $D_8 = \{1, 2, 4, 8\}$



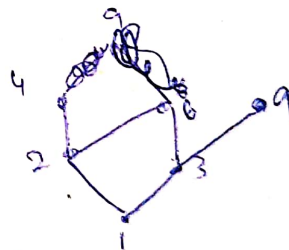
ii) $D_6 = \{1, 2, 3, 6\}$



iii)

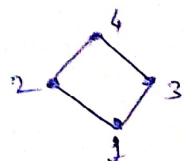
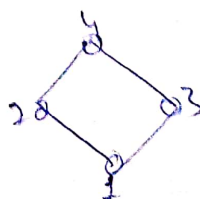
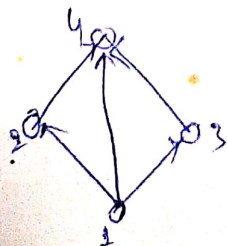
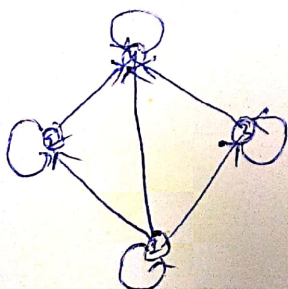


iv)



Ex:

Determine the Hasse diagram of the relation $R, A = \{1, 2, 3, 4\}$
 $R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$



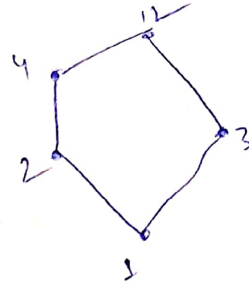
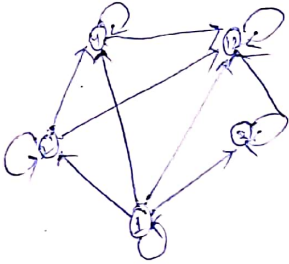
Well-Ordered Set :

Ex: Draw Hasse diagram for the following relations on set

(10) (C)

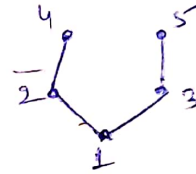
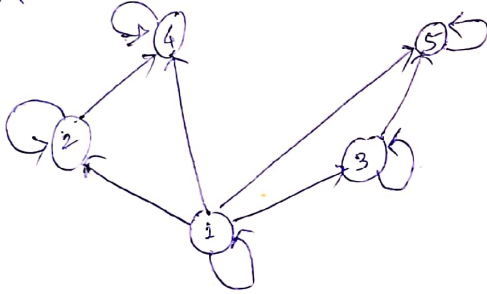
i) $A = \{1, 2, 3, 4, 12\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (12,12), (1,2), (4,12), (1,3), (1,4), (1,12), (2,4), (2,12), (3,12)\}$



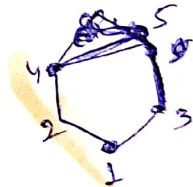
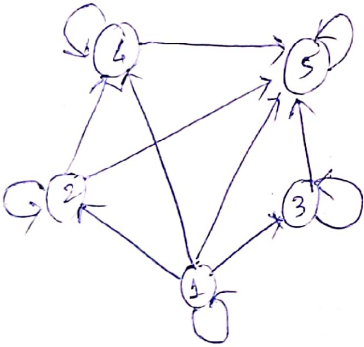
ii) $A = \{1, 2, 3, 4, 5\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,4), (3,5), (2,2), (3,3), (4,4), (5,5)\}$



iii) $A = \{1, 2, 3, 4, 5\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (4,5), (1,3), (1,4), (1,5), (2,4), (2,5), (3,5)\}$



4. Determine the Hasse diagram of the relation $A = \{1, 2, 3, 4, 5\}$, whose matrix is shown

i)

$M_R =$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

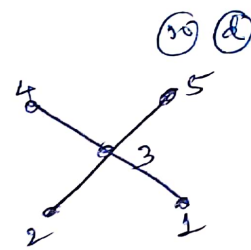
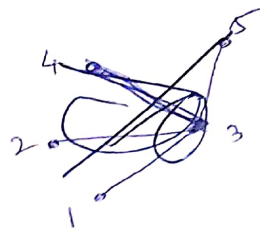


Well-Ordered

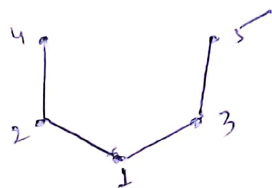
A set with an ordering relation

Ex) $M_R =$

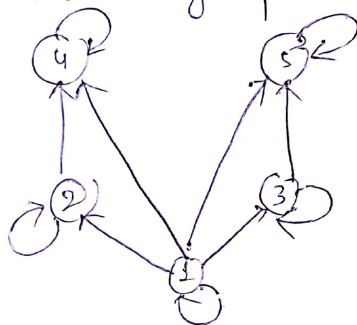
	1	2	3	4	5
1	1	0	1	1	1
2	0	1	1	1	1
3	0	0	1	1	1
4	0	0	0	1	0
5	0	0	0	0	1



Ex. Determine the matrix of the partial order whose diagram is



Q. Make digraph



$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,4), (1,4), (1,3), (3,5), (1,5)\}$$

Matrix

$$M_R =$$

	1	2	3	4	5
1	1	1	1	1	1
2	0	1	0	1	0
3	0	0	1	0	1
4	0	0	0	1	0
5	0	0	0	0	1

A set with an ordering relation is well-order if every

Combination of Partial Ordered Sets or Components of Poset.

(11)

Let $(P, \leq)$ be a poset.

An element $a \in P$ is the greatest element of P if $x \leq a$ for all $x \in P$ i.e., every element in P precedes a . The greatest element, if it exists, is unique.

Similarly, an element $b \in P$ is called the least element if $b \leq x$ for all $x \in P$ i.e., every element in P succeeds b . The least element is unique if it exists.

An element 'a' in the poset is called a maximal element of P if $a \not\leq x$ for no x in P , that is, if no element of P strictly succeeds 'a'.

Similarly an element 'b' in P is called a minimal element of P if $x \not\leq b$ for no x in P .

Maximal and minimal elements ~~are~~ are easy to spot using a Hasse diagram. They are the top and bottom elements in the diagram. That is, a maximal element has no connections leading up and a minimal element has no connections leading down. A greatest element is connected to every other element by a path leading down and a least element is connected to every other element by a path leading up.

The following points are to be noted:

- A poset may not have a maximal element. For ex, the natural numbers under usual $\leq$ have no maximal element.
- A poset may have more than one maximal or minimal element. In the poset $\{2, 3, 4, 6\}$ under divisibility, 4 and 6 are both maximal elements.
- Maximal element may not be the greatest element. In above 4 and 6 are maximal but neither 4 nor 6 is the greatest element.

(21)

A set with an ordering relation is well-order if every

d) A poset may have a maximal element but no minimal elements, or a minimal element but no maximal elements. For ex., the poset $(\mathbb{Z}^-, \leq)$ has a maximal element but no minimal elements, whereas the poset $(\mathbb{Z}^+, \leq)$ has a minimal element but no maximal elements.

e) In a totally ordered set the concepts of minimal and least coincide as do those of maximal and greatest.

Ex: Let $S = \{a, b, c\}$. Then $(P(S), \subseteq)$ is a poset.

a) Let $A = \{\emptyset, \{b\}, \{c\}, \{a, c\}\}$

Then $(A, \subseteq)$ is a poset with $\emptyset$ as least element and A has no greatest element.

b) Let $B = \{\{b\}, \{c\}, \{a, b\}, \{a, b, c\}\}$

Then $(B, \subseteq)$ is a poset with $\{a, b, c\}$ as greatest element and B has no least element.

c) Let $C = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Then $(C, \subseteq)$ is a poset with $\emptyset$ as least and $\{a, b\}$ as greatest element.

Ex: Find the least and greatest element in the poset $(\mathbb{Z}^+, |)$ if they exist.

→ The least element of the poset $(\mathbb{Z}^+, |)$ is 1 since $1/n$ whenever n is a positive integer. There is no greatest element since there is no integer which is divisible by all positive integers.

Well-Ordered Set :

(14)

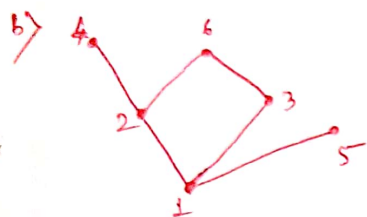
A set with an ordering relation is well-ordered if .

Ex: Determine whether the posets represented by Hasse diagrams in the figure below have a greatest element, least element, minimal element and maximal elements.

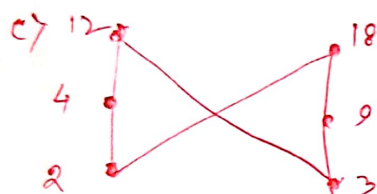
(13)



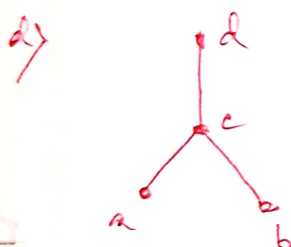
The least element of the poset is 1 and is the only minimal element. The greatest element is 27 and is the only maximal element.



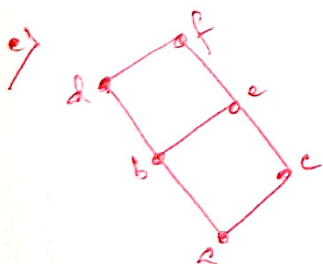
Least element = 1 Minimal element = 1
Greatest element = no Maximal element = 4, 5, 6.



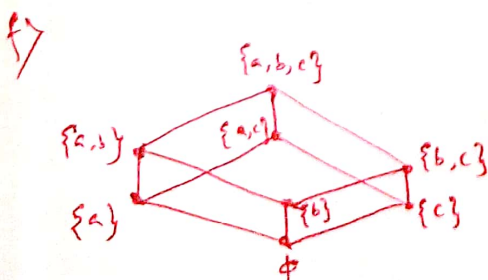
Least element = no Minimal element = 2, 3
Greatest element = no Maximal element = 12, 18



Least element = no Minimal element = a, b
Greatest element = d Maximal element = d.



Least element = a Minimal element = a
Greatest element = f Maximal element = f.



Least element = ϕ Minimal element = ϕ
Greatest element = {a, b, c} Maximal element = {a, b, c}

(19)

(3)

Well-Ordered Set :

A set with an ordering relation is well-ordered if every non-empty subset of the set has a least element.

For ex, the set of natural numbers is well-ordered.

The set of integers $\mathbb{Z}$ is not well-ordered since the set of negative integers, which is a subset of $\mathbb{Z}$, has no least element.

Upper and Lower bound :

Let B be a subset of a poset $(A, \leq)$. An element $u \in A$ is called an upper bound of B if u succeeds every element of B i.e., $x \leq u$ for all $x \in B$. An element $l \in A$ is called a lower bound of B if l precedes every element of B i.e., $l \leq x$ for all $x \in B$.

An element $a \in A$ is called a least upper bound (lub) of B if ' a ' is an upper bound of B and $a \leq a'$, whenever a' is the upper bound of B . Thus $a = \text{lub}(B)$.

Similarly, an element $a \in A$ is called the greatest lower bound (glb) of B if ' a ' is a lower bound of B and $a' \leq a$, whenever a' is a lower bound of B . Thus $a = \text{glb}(B)$.

Note :

Some texts use the term supremum of A instead of the term least upper bound and write $\sup(A)$ instead of $\text{lub}(A)$ and use the term infimum of A instead of the term greatest lower bound and write $\inf(A)$ instead of $\text{glb}(A)$.

Consider the poset $A = (\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, \mid)$. Find the lower bound and least ...

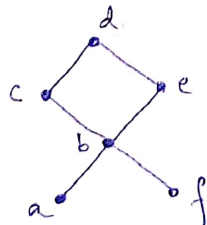
- 2) A subset B of a poset may or may not have upper or lower bounds.
- 3) An upper or lower bound may or may not belong to B itself.
- 4) There can be more than one upper bound and lower bound of a set.
- 5) The greatest element is always the supremum but the converse is not true. In fact $a = \sup(B)$ is the greatest element iff $a \in B$.
- 6) The least element is always the infimum but the converse is not true. In fact $b = \inf(B)$ is the least element iff $b \in B$.

Ex: In the poset $A = (\{1, 2, 3, 4, \dots, 10\}, \mid)$ the subset $\{2, 7\}$ has no upper bound since there is no integer in A which is divisible by both 2 and 7. The lower bound of the subset is 1 since 2 and 7 are divisible by only 1. Hence, 1 is the greatest lower bound for $\{2, 7\}$ i.e., $\text{glb } \{2, 7\} = 1$.

The subset $\{1, 2, 3\}$ has 6 and 1 as unique upper and lower bounds. Hence $\text{lub } \{1, 2, 3\} = 6$ and $\text{glb } \{1, 2, 3\} = 1$.

The subset $\{1, 2, 4\}$ has 4 and 8 as upper bounds. Hence $\text{lub } \{1, 2, 4\} = 4$.

Ex: Consider the poset whose diagram is given by



i) Upper bound for $\{c, e\}$ is d
 $\text{lub } \{c, e\}$ or $\sup \{c, e\} = d$.

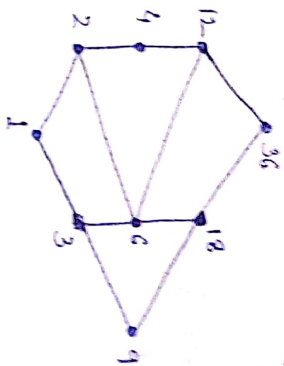
Lower bound for $\{c, e\}$ are b, a and f
 $\text{glb } \{c, e\}$ or $\inf \{c, e\} = b$.

ii) Upper bound for $\{a, f\}$ are b, c, e, d
 $\text{lub } \{a, f\}$ or $\sup \{a, f\} = b$.

Lower bound for $\{a, f\}$ do not exist.

Ex: Consider the poset $A = (\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, \preceq)$, find the greatest lower bound and least upper bound of the sets $\{6, 18\}$ and $\{4, 6, 9\}$.

→ Hasse diagram under relation of divisibility



An integer is a lower bound of $\{6, 18\}$ if 6 and 18 are divisible by that integer. Only such integers are 1 and 6. Since 16, 6 is the greatest lower bound of $\{6, 18\}$.

Hence $\text{glb}\{6, 18\} = 6$.

The integer is an upper bound of $\{6, 18\}$ if and only if it is divisible by 6 and 18 which is 18 and 36.

Hence $\text{lub}\{6, 18\} = 18$.

The only lower bound of $\{4, 6, 9\} = 1$. Hence $\text{glb}\{4, 6, 9\} = 1$.

Note: $\text{glb}\{a, b\} = \text{glb}\{a, b\}$ and $\text{lub}\{a, b\} = \text{lub}\{a, b\}$. Hence $\text{lub}\{4, 6, 9\} = 36$.

Chain: Let A be a set that is partially ordered with respect to a relation. A subset B of A is called a chain if and only if each pair of elements in B is comparable. Thus, if A is a chain and $x, y \in A$ then either $x \leq y$ or $y \leq x$.

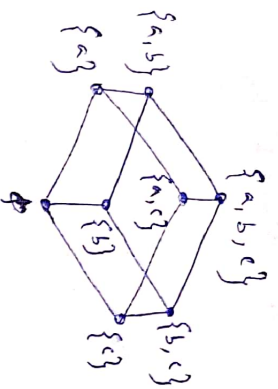
The length of a chain is one less than the number of elements in the chain, i.e., length is the number of links that the chain has.

A maximal chain is a chain that is not a subset of a larger chain.

A subset of a poset is called an antichain if every two elements of this subset are incomparable.

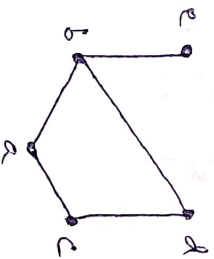
then or equal to " or integers. Show that $(A, \leq)$ is a poset.

Ex: Consider the poset $[P(\{a, b, c\}), \subseteq]$. Draw the Hasse diagram and find a chain of length 3 in the poset.



Since $\emptyset \subseteq \{a\} \subseteq \{a, b\} \subseteq \{a, b, c\}$, the set $S = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}\}$ is a chain of length 3 in $[P(\{a, b, c\}), \subseteq]$. There are total six chains of length 3.

Ex: Find the antichain with the greatest number of elements in the poset with the following Hasse diagram



The only antichain with more than one element are $\{b, c\}$, $\{c, d\}$ and $\{a, c\}$.

Isomorphism: Let $(A, \leq)$ and $(B, \leq)$ be posets. A one-to-one map $f: A \rightarrow B$ is called an isomorphism from $(A, \leq)$ to $(B, \leq)$ if $x \leq y \Leftrightarrow f(x) \leq f(y)$ for all $x, y \in A$. The posets $(A, \leq)$ and $(B, \leq)$ are isomorphic posets.

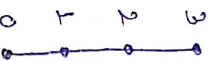
These diagrams of both the posets are identical except for the labeling of vertices.

Ex: Let $A = \{1, 2, 4, 8\}$ and let $\leq$ be the partial order of divisibility on A . Let $A' = \{0, 1, 2, 3\}$ and let $\leq$ be the usual relation "less than or equal to" on integers. Show that $(A, \leq)$ and $(A', \leq)$ are isomorphic posets.

→ The Hasse diagram of $(A, \leq)$ is



and the Hasse diagram of $(A', \leq)$ is



The mapping $f: A \rightarrow A'$.

Since both have same Hasse diagram, every element of A map to only a single element in A' and both have same number of elements. Hence it is one-one onto.

If we define $f(1) = 0$, $f(2) = 1$, $f(4) = 2$, $f(8) = 3$.

At a/b iff $f(a) \leq f(b)$

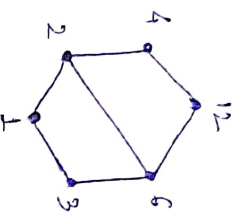
$\therefore 1/2 \Rightarrow f(1) < f(2)$
 $\Rightarrow 0 < 1$

and $2/4 \Rightarrow f(2) < f(4)$
 $\Rightarrow 1 < 2$

Hence $f: A \rightarrow A'$ is an isomorphism.

Ex: Let $A = \{1, 2, 3, 4, 6, 12\}$. Let K be the partial order relation " $\leq$ " on A given by $x \leq y$ iff $x \mid y$. Then $(A, \leq)$ is a poset. Consider another poset $(A, \leq)$ in which $\leq$ denotes the usual "less than equal to" relation on A . Then $(A, \leq)$ and $(A, \leq)$ are not isomorphic.

The Hasse diagrams of $(A, \leq)$ and $(A, \leq)$ are given below



The poset $(A, \leq)$ and $(A, \leq)$ are not isomorphic. Let 4 and 6 be two distinct elements in $(A, \leq)$ and $f(4), f(6)$ be two distinct elements in $(A, \leq)$. Now $f(4)$ and $f(6)$ are comparable in $(A, \leq)$, while 4 and 6 are non-comparable.

Lattice: A poset $(P, \leq)$ is called a lattice if every 2-element subset of P has both a least upper bound and a greatest lower bound, i.e., if $\text{lub}(x, y)$ and $\text{glb}(x, y)$ exist for every x and y in P . In this case, we denote

$$x \vee y = \text{lub}\{x, y\} \quad (\text{read as join of } x \text{ and } y)$$

$$x \wedge y = \text{glb}\{x, y\} \quad (\text{read as meet of } x \text{ and } y)$$

Note:

Every chain is a lattice. Since any two elements x, y of a chain are comparable, we find

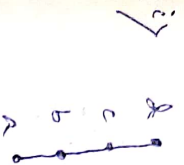
$$x \vee y = \text{lub}\{x, y\} = y$$

$$x \wedge y = \text{glb}\{x, y\} = x$$

2) Lattice is a mathematical structure equipped with two binary operations join and meet.

3) Other notations of join are $\oplus$ or $+$ and meet are $\times$ and $\cdot$.

Ex: Determine whether the following Hasse diagrams represent lattice or not.

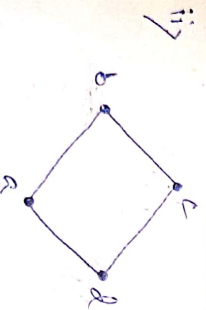


→ set is constant the closure tables for $\text{lub}(V)$ and $\text{glb}(V)$.

V	a	b	c	d
a	a	a	c	d
b	b	b	b	d
c	c	c	c	d
d	d	d	d	d

$\wedge$	a	b	c	d
a	a	a	a	a
b	b	a	b	b
c	c	a	b	c
d	d	a	b	c

Since each subset of two elements has least upper bound and a greatest lower bound, so this is the lattice.



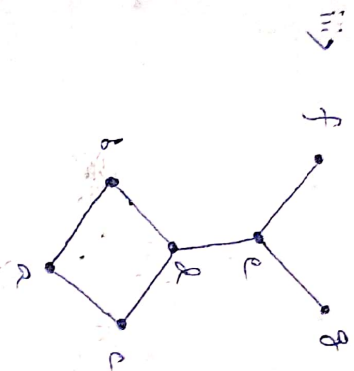
→ $\text{lub}(V)$:

V	a	b	c	d
a	a	a	c	d
b	b	b	b	c
c	c	c	c	c
d	d	d	c	d

$\text{glb}(V)$:

2	2	5	2	2
2	2	2	2	2
2	5	5	2	2
2	2	5	2	2
2	2	2	2	2

Since each subset of two elements has least upper bound and a greatest lower bound, so this is the lattice.



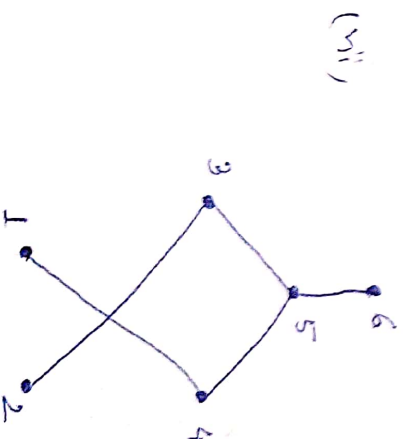
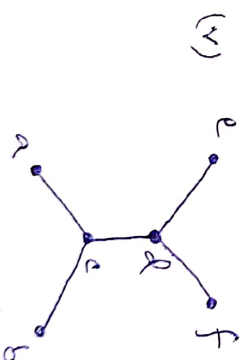
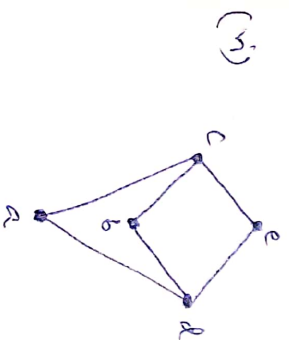
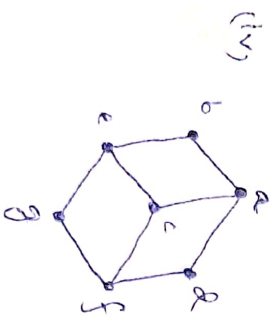
→
 $\text{sub}(V):$

V	a	b	c	d	e	f	g
a	a	b	c	d	e	f	g
b	a	b	c	d	e	f	g
c	a	b	c	d	e	f	g
d	a	b	c	d	e	f	g
e	a	b	c	d	e	f	g
f	a	b	c	d	e	f	g
g	a	b	c	d	e	f	g

$\text{sub}(V):$

V	a	b	c	d	e	f	g
a	a	a	a	a	a	a	a
b	a	b	a	b	b	b	b
c	a	a	c	c	c	c	c
d	a	b	c	d	d	d	d
e	a	b	c	d	e	e	e
f	a	b	c	d	e	f	f
g	a	b	c	d	e	f	g

Since each subset of two elements has no least upper bound but has greatest lower bound. So this is not the lattice.



Properties of Lattices:

① If L be any lattice, then for any $a, b, c \in L$

$$1) a \vee a = a \quad b) a \wedge a = a \quad (\text{Idempotency})$$

$$2) a) a \vee b = b \vee a \quad b) a \wedge b = b \wedge a \quad (\text{Commutativity})$$

$$3) a) a \vee (b \vee c) = (a \vee b) \vee c \quad b) a \wedge (b \wedge c) = (a \wedge b) \wedge c \quad (\text{Associativity})$$

$$4) a) a \vee (a \wedge b) = a \quad b) a \wedge (a \vee b) = a \quad (\text{Absorption})$$

Proof:

$$1) a) a \vee a = \inf\{a, a\} = \inf\{a\} = a$$

$$b) a \wedge a = \sup\{a, a\} = \sup\{a\} = a$$

$$2) a) a \vee b = \inf\{a, b\} = \inf\{b, a\} = b \vee a$$

$$b) a \wedge b = \sup\{a, b\} = \sup\{b, a\} = b \wedge a$$

$$3) a) \text{ We have } (a \vee b) \vee c = \inf\{a \vee b, c\}.$$

$$\text{So, } a \vee b \leq (a \vee b) \vee c \\ \text{and } c \leq (a \vee b) \vee c.$$

$$\text{Also, } a \leq a \vee b \text{ and } b \leq a \vee b$$

$$\text{Hence, } a \leq (a \vee b) \vee c \text{ and } b \leq (a \vee b) \vee c \text{ by transitivity.}$$

Thus, $(a \vee b) \vee c$ is an upper bound of a and b .

By definition of $\inf$ of b and c i.e. $b \vee c$.

$$a, b \vee c \leq (a \vee b) \vee c.$$

So it is upper bound of a and $b \vee c$. Thus by definition of $\vee$, we have

$$a \vee (b \vee c) \leq (a \vee b) \vee c.$$

$$\text{Similarly, } (a \vee b) \vee c \leq a \vee (b \vee c).$$

$$(a \vee b) \vee c = a \vee (b \vee c)$$

By symmetry of $\leq$ we see that

the proof of part (a).

4) a) since $a \vee b \leq a$ and $a \leq a$, a is an upper bound of $a \vee b$ and a , so $a \vee (a \vee b) \leq a$. By definition of lub, we have,

$a \leq a \vee (a \vee b)$. Hence by antisymmetry of $\leq$, we see that
 $a \vee (a \vee b) = a$.

b) The proof is similar to the proof of (a).

⑧ If L be a lattice, then for every a and b in L

- a) $a \vee b = b$ if and only if $a \leq b$
- b) $a \wedge b = a$ if and only if $a \leq b$
- c) $a \wedge b = a$ if and only if $a \vee b = b$.

Proof:

a) Let $a \vee b = b$.

Since $a \leq a \vee b = b$, we get $a \leq b$.

Conversely, if $a \leq b$, then, since $b \leq b$, b is an upper bound of a and b .

So, by definition of lub, we have, $a \vee b \leq b$.

Again, since $a \vee b$ is an upper bound, $b \leq a \vee b$, so $a \vee b = b$.

b) Similar to (a)

c) Combining (a) and (b), we get (c).

⑨ Let $(L, \leq)$ be a lattice. Then for $a, b, c, d \in L$,

- 1) $a \leq b \Rightarrow a \vee c \leq b \vee c$
- 2) $a \leq b \Rightarrow a \wedge c \leq b \wedge c$
- 3) $a \leq b$ and $c \leq d \Rightarrow a \vee c \leq b \vee d$
- 4) $a \leq b$ and $c \leq d \Rightarrow a \wedge c \leq b \wedge d$.

Proof:

1) From the property of lattices, we have,
 $a \leq b \Leftrightarrow a \vee b = b.$ ————— (i)

Now, we get,

$$\begin{aligned}
 (a \vee c) \vee (b \vee c) &= (a \vee c) \vee (c \vee b) && \text{(Commutative)} \\
 &= a \vee (c \vee c) \vee b && \text{(Associative)} \\
 &= a \vee (c \vee b) && (\because c \vee c = c) \\
 &= a \vee (b \vee c) && \text{(Commutative)} \\
 &= (a \vee b) \vee c && \text{(Associative)} \\
 &= b \vee c && \text{(From (i))}
 \end{aligned}$$

Thus, by above property,

$$a \vee c \leq b \vee c.$$

2) Proof is similar as (1).

3) From the property of lattices, we have,

$$a \leq b \Leftrightarrow a \vee b = b \quad \text{and} \quad c \leq d \Leftrightarrow c \vee d = d. \quad \text{————— (ii)}$$

Now,

$$\begin{aligned}
 (a \vee c) \vee (b \vee d) &= a \vee (c \vee b) \vee d && \text{(Associative)} \\
 &= a \vee (b \vee c) \vee d && \text{(Commutative)} \\
 &= (a \vee b) \vee (c \vee d) && \text{(Associative)} \\
 &= b \vee d && \text{(From (ii))}
 \end{aligned}$$

Thus, by above property,

$$a \vee c \leq b \vee d.$$

4) Proof is similar as (3).

(2) In any lattice the distributive inequalities

$$i) a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c)$$

$$ii) a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c) \quad \text{hold for any } a, b, c.$$

Proof:

$$i) \quad a \wedge b \leq a$$

$$a \wedge b \leq b \leq b \vee c$$

$$\Rightarrow a \wedge b \text{ is lower bound of } \{a, b \vee c\}$$

$$\Rightarrow a \wedge b \leq a \wedge (b \vee c) \quad \text{--- (1)}$$

$$\text{Again,} \quad a \wedge c \leq a$$

$$a \wedge c \leq c \leq b \vee c$$

$$\Rightarrow a \wedge c \leq a \wedge (b \vee c) \quad \text{--- (2)}$$

(1) and (2) show that $a \wedge (b \vee c)$ is an upper bound of $\{a \wedge b, a \wedge c\}$

$$\Rightarrow (a \wedge b) \vee (a \wedge c) \leq a \wedge (b \vee c)$$

$$\text{Hence,} \quad a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c)$$

ii) The proof is similar to (i).

Principle of Duality:

We observe that if $\leq$ is a partial order on any set, then its inverse relation $\geq$ is also a partial order. Also it follows from the definitions, that the dual of a and b with respect to $\leq$ is the same as the glb with respect to the relation $\geq$ and vice versa. This observation can be developed into a formal principle of duality for lattices, which says that if we interchange $\vee$ with $\wedge$ and $\leq$ with $\geq$ in a true statement about lattices, we get another true statement. And the corresponding statements are called dual of each other.

Product of lattices:

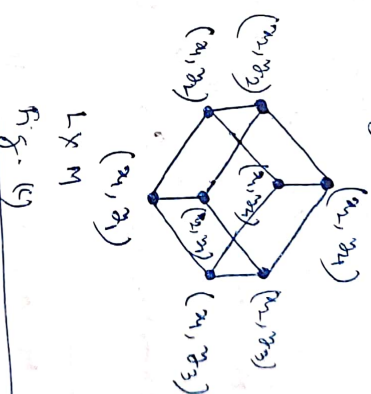
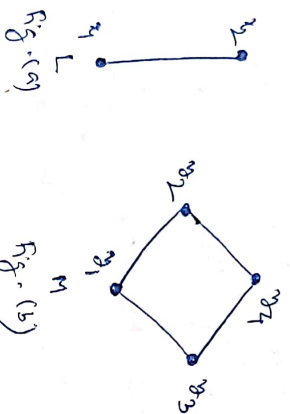
Let L and M be lattices. The set of ordered pairs $\{(x, y) : x \in L, y \in M\}$ with operations $\vee$ and $\wedge$ defined by

$$(x_1, y_1) \vee (x_2, y_2) = (x_1 \vee x_2, y_1 \vee y_2)$$

$$(x_1, y_1) \wedge (x_2, y_2) = (x_1 \wedge x_2, y_1 \wedge y_2)$$

is the direct product of L and M are denoted by $L \times M$. It is also called the product partial order.

The direct product of the lattices L and M can be graphically be described in terms of the Hasse diagram.



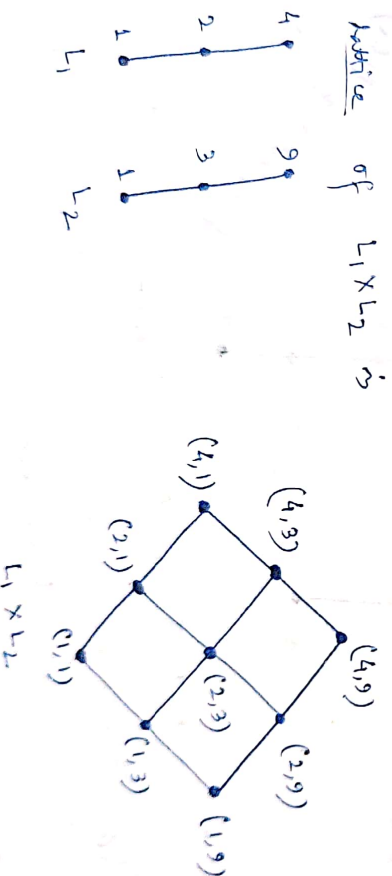
Ex: Let L and M be two lattices as shown in Fig. (a) and (b) respectively

Then, $L \times M$ is the lattice shown in Fig. (c)

Ex: $L_1 = \{1, 2, 4\}$, $L_2 = \{1, 3, 9\}$ and $a \leq b$ if a divides b . Draw the lattice $L_1 \times L_2$

$\rightarrow L_1 \times L_2 = \{(1,1), (1,3), (1,9), (2,1), (2,3), (2,9), (4,1), (4,3), (4,9)\}$

The lattice of $L_1 \times L_2$ is



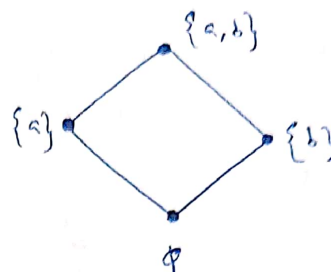
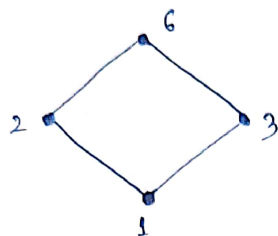
Ex: Let L_1 be the lattice D_6 (divisor of 6) = $\{1, 2, 3, 6\}$ and let L_2 be the lattice $(P(S), \subseteq)$ where $S = \{a, b\}$. The two lattices are isomorphic.

→ The Hasse diagram of two lattices L_1 and L_2 are given below.

Let $f: L_1 \rightarrow L_2$ is defined by

$$f(1) = \phi, f(2) = \{a\}, f(3) = \{b\}, f(6) = \{a, b\}$$

Then, clearly, f is one-one function



now, consider, $f(2 \wedge 3) = f(1) = \phi$

$$f(2) \wedge f(3) = \{a\} \wedge \{b\} = \phi.$$

and $f(2 \vee 3) = f(6) = \{a, b\}$

$$f(2) \vee f(3) = \{a\} \vee \{b\} = \{a, b\}.$$

Since, the mapping is one-one and onto f is isomorphic and hence the lattices are isomorphic.

Sublattices: A non empty subset L' of a lattice L is called a sublattice of L if $a, b \in L' \Rightarrow a \vee b, a \wedge b \in L'$ i.e., the algebra $(L', \vee, \wedge)$ is a sublattice of $(L, \wedge, \vee)$ iff L' is closed under both operations $\wedge$ and $\vee$.

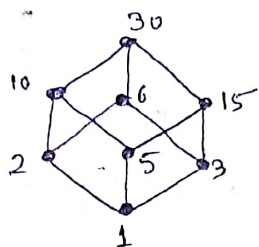
Note: From the definition it follows that a sublattice itself is a lattice and every singleton of a lattice L is a sublattice of L . However, any subset of L , which is a lattice need not be a sublattice.

Ex: If S be a set and $L = (P(S), \cap, \cup)$. Then only those subcollections of $P(S)$ are sublattices of L which are closed under $\cap$ and $\cup$. If $S = \{a, b, c, d\}$, then $\{\phi, \{a\}, \{a, c\}, \{c\}, \{a, b, c\}\}$ would be sublattice but $\{\phi, \{a\}, \{b\}, \{c\}, \{b, c\}\}$ would not form a sublattice as $\{a\} \cup \{c\} = \{a, c\}$ is not there. ⁽¹⁰⁾
 as every element in L has a complement. ⁽¹¹⁾

Ex: Show with an example that the union of two sublattices⁽²¹⁾ may not be a sublattice.

→ Consider the lattice $L = \{1, 2, 3, 4, 6, 12\}$ of factors of 12 under divisibility. Then $S = \{1, 2\}$ and $T = \{1, 3\}$ are sublattices of L . But the union of S and T i.e., $S \cup T = \{1, 2, 3\}$ is not a sublattice since $2, 3 \in S \cup T$ but $2 \vee 3 = 6 \notin S \cup T$.

Ex: Let D_n denote the set of all positive divisors of n . If $L = D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ having the Hasse diagram shown in the figure below.



then $D_6 = \{1, 2, 3, 6\}$, $D_{10} = \{1, 2, 5, 10\}$, $D_{15} = \{1, 3, 5, 15\}$ are sublattices. Another sublattice is $\{5, 10, 15, 30\}$. In general if $m|n$ then D_m is a sublattice of D_n . Also all the multiples of m in D_n form a sublattice.

Note: A lattice may fail to possess a glb or lub for arbitrary subset.

Complete lattice: A lattice is called complete if each of its non empty subsets has a least upper bound and a greatest lower bound.

Every finite lattice is complete because every subset here is finite. Also every complete lattice must have a least element and a greatest element. The least and the greatest elements of a lattice are called bounds (units, universal bounds) of the lattice and are denoted by 0 and 1 respectively.

A lattice which has both elements 0 and 1 is called a bounded lattice.

Note:

- 1) Every finite lattice $(L, \vee, \wedge)$ with $L_n = \{a_1, a_2, \dots, a_n\}$ is bounded. Here $\bigwedge_{i=1}^n a_i = 0$ and $\bigvee_{i=1}^n a_i = 1$.
- 2) $(P(S), \cap, \cup)$ is bounded with $\phi = 0$ and $S = 1$.
- 3) In a bounded lattice, the bounds 0 and 1 clearly satisfy:

$$\begin{aligned} 0 \wedge a &= 0 = a \wedge 0 && \text{for all } a \in L \\ 0 \vee a &= a = a \vee 0 && \text{for all } a \in L \\ 1 \wedge a &= a = a \wedge 1 && \text{for all } a \in L \\ 1 \vee a &= 1 = a \vee 1 && \text{for all } a \in L \end{aligned}$$

Complement of an element in a lattice:

Defⁿ: In a bounded lattice $(L, \vee, \wedge, 0, 1)$, an element $b \in L$ is called a complement of an element $a \in L$ if

$$a \wedge b = 0 \quad \text{and} \quad a \vee b = 1$$

where 0 and 1 are lower and upper bound of L.

Note:

- 1) The definition of a complement is symmetric in a and b, so that b is complement of a if a is a complement in b.
 - 2) Any element $a \in L$ may or may not have a complement.
 - 3) ~~Any~~ An element $a \in L$ may have more than one complement in L, so complements are not unique.
 - 4) In any bounded lattice, the bound 0 and 1 are unique complements of each other because $0 \vee 1 = 1$ and $0 \wedge 1 = 0$.
- A lattice L is called complemented lattice if it is bounded and if every element in L has a complement.

Ex: In fig (a), complement of a_1 is a_2 since $a_1 \wedge a_2 = 0$ and $a_1 \vee a_2 = 1$. In fig (b), both a_1 and a_2 are complements of a_3 . Since $a_1 \wedge a_3 = 0$, $a_2 \wedge a_3 = 0$ and $a_1 \vee a_3 = 1$, $a_2 \vee a_3 = 1$ and so here complements are not unique.

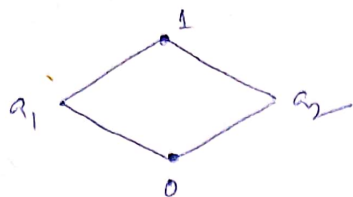


fig (a)

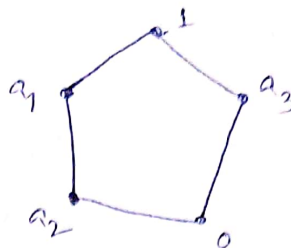
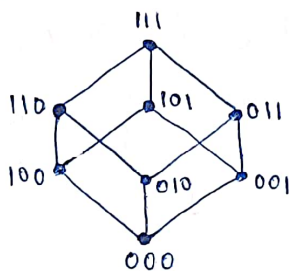


fig. (b)

Ex: Show that the lattice $(L^3, \leq_3)$ of 3 tuples of 0 and 1 is complemented.

→ The Hasse diagram of the lattice $(L^3, \leq_3)$ is shown in the figure below.



The bounds of the lattice are $(0,0,0)$ and $(1,1,1)$. This is a complemented lattice in which every element has a unique complement.

Note: In general, the lattice $(L^n, \leq_n)$ is a complemented lattice.

The complement of an element of L^n can be obtained by interchanging 1 by 0 and 0 by 1.

Distributive lattice: A lattice $(L, \vee, \wedge)$ is called a distributive lattice if for any $a, b, c \in L$

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Note: Both the equalities are equivalent to one another, hence to check whether the lattice is distributive or not, it is sufficient to verify one of them.

Ex: $L = (P(S), \cup, \cap)$ is distributive lattice, since the distributive laws for $\cup$ over $\cap$ and $\cap$ over $\cup$ are well known facts of set theory. (32)

Ex: Show that the lattices given in the following diagrams are not distributive.

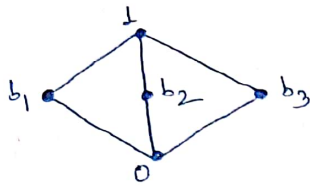


Fig. (a) Diamond

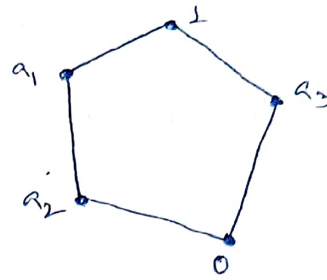


Fig. (b) Pentagon

→ In the lattice shown in (a)

$$b_1 \wedge (b_2 \vee b_3) = b_1 \wedge 1 = b_1$$

$$\text{but } (b_1 \wedge b_2) \vee (b_1 \wedge b_3) = 0 \vee 0 = 0.$$

Hence the lattice shown in (a) is not distributive.

In the lattice shown in (b)

$$a_1 \wedge (a_2 \vee a_3) = a_1 \wedge 1 = a_1$$

$$\text{but } (a_1 \wedge a_2) \vee (a_1 \wedge a_3) = a_2 \vee 0 = a_2 \neq a_1$$

Hence the lattice shown in (b) is not distributive.

Note:

- (1) Observe that the distributive equalities may be satisfied for a particular combination in a particular order but this does not guarantee that the lattice is distributive.

For ex, in the diagram (b),

$$a_3 \wedge (a_1 \vee a_2) = 0 = (a_3 \wedge a_1) \vee (a_3 \wedge a_2).$$

- (2) In particular if any lattice contains at least one of these five elements lattices as a sublattice, then it cannot be distributive. These are the smallest non-distributive lattices.

(3) In other words, a lattice $(L, \wedge, \vee)$ is distributive if and only if it does not contain the five element pentagonal or, the diamond lattice given above as one of its sublattices (or an isomorphic copy thereof).

Theorem: In a distributive lattice $(L, \leq)$

$a \wedge b = a \wedge c$ and $a \vee b = a \vee c$ imply that $b = c$.

Proof:

$$\begin{aligned}
 b &= b \wedge (a \vee b) && \text{by absorption} \\
 &= b \wedge (a \vee c) && [\because a \vee b = a \vee c] \\
 &= (b \wedge a) \vee (b \wedge c) && \text{by distributivity} \\
 &= (a \wedge b) \vee (b \wedge c) && \text{by commutativity} \\
 &= (a \wedge c) \vee (b \wedge c) && [\because a \wedge b = a \wedge c] \\
 &= (c \wedge a) \vee (c \wedge b) && \text{by commutativity} \\
 &= c \wedge (a \vee b) && \text{by distributivity} \\
 &= c \wedge (a \vee c) && [\because a \vee b = a \vee c] \\
 &= c && \text{by absorption} \\
 &&& \text{(proved)}
 \end{aligned}$$

Theorem: In a distributive lattice, if an element has a complement then this complement is unique.

Proof: Suppose that an element a has two complements b and c .

$$\begin{aligned}
 \text{Then} \quad a \vee b &= 1 & a \wedge b &= 0 \\
 a \vee c &= 1 & a \wedge c &= 0
 \end{aligned}$$

We have,

$$\begin{aligned}
 b &= b \wedge 1 \\
 &= b \wedge (a \vee c) && [\because a \vee c = 1] \\
 &= (b \wedge a) \vee (b \wedge c) && \text{by distributivity} \\
 &= 0 \vee (b \wedge c) \\
 &= (a \wedge c) \vee (b \wedge c) && [\because a \wedge c = 0] \\
 &= (a \vee b) \wedge c \\
 &= 1 \wedge c && [\because a \vee b = 1] \\
 &= c
 \end{aligned}$$

Hence the complements are unique. (proved)

Theorem: Show that every chain is a distributive lattice. (34)

Proof: Let $(L, \leq)$ be a chain and $a, b, c \in L$. Since L is a chain, either $a \leq b$ or $b \leq a$. If $a \leq b$, then $a \vee b = b$ and $a \wedge b = a$. Hence, for any two elements $a, b \in L$, $a \wedge b$ and $a \vee b$ exist in L . Suppose $a \leq b$.

Case 1: $b \leq c$

$$\text{Now, } a \wedge (b \vee c) = a \wedge c = a$$

$$\text{and } (a \wedge b) \vee (a \wedge c) = a \vee a = a$$

$$\text{Hence, we have, } a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Case 2: $c \leq b$

Suppose $a \leq c$

In this case, we have, $a \leq c \leq b$.

$$\text{Now, } a \wedge (b \vee c) = a \wedge b = a$$

$$\text{and } (a \wedge b) \vee (a \wedge c) = a \vee a = a$$

$$\text{Hence, } a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Similarly, if $b \leq a$, then $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

Hence the theorem. (proved)

Modular Lattice: A lattice $(L, \leq)$ is said to be modular if $a \vee (b \wedge c) = (a \vee b) \wedge c$ whenever $a \leq c$ for all $a, b, c \in L$.

Theorem: Every distributive lattice is modular.

Proof: Let $(L, \leq)$ be a distributive lattice and $a, b, c \in L$ be such that $a \leq c$. Thus if $a \leq c$ then $a \vee c = c$.

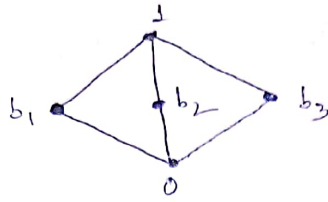
$$\text{Now, } a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = (a \vee b) \wedge c.$$

Hence, every distributive lattice is modular.

Ex: ① Any chain is modular. (13)

② The converse is not true, ~~and~~ is, every modular ⁽³⁵⁾ lattice is distributive lattice, as shown by the following example.

Ex: Prove that the lattice given by the following diagram is modular.



→ We know that the diamond lattice is nondistributive. Since the lattice is symmetric with respect to b_1, b_2, b_3 , the only situations are $b_1 < 1$ and $0 < b_1$ when $a < c$.

Thus taking $a = b_1$ and $c = 1$.

$$a \vee (b \wedge c) = b_1 \vee (b \wedge 1) = b_1 \vee b$$

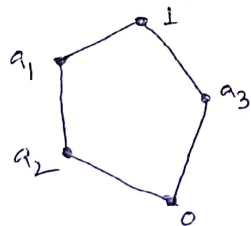
$$\text{and } (a \vee b) \wedge c = (b_1 \vee b) \wedge 1 = b_1 \vee b, \text{ whatever be } b.$$

Similarly, for $a = 0, c = b_1$,

$$a \vee (b \wedge c) = 0 \vee (b \wedge b_1) = b \wedge b_1$$

$$\text{while } (a \vee b) \wedge c = (0 \vee b) \wedge b_1 = b \wedge b_1 \text{ and so modularity holds.}$$

Ex: Prove that the pentagonal lattice is not modular.



Proof:
We take $a = a_2, b = a_3, c = a_1$ then $a < c$.
But $a \vee (b \wedge c) = a_2 \vee (a_3 \wedge a_1) = a_2 \vee 0 = a_2$
whereas $(a \vee b) \wedge c = (a_2 \vee a_3) \wedge a_1 = 1 \wedge a_1 = a_1$ and these are unequal.
Hence (proved).

Note: A lattice $(A, \wedge, \vee)$ is modular if and only if it does not have a sublattice isomorphic with the pentagonal lattice.

theorem: If $(L, \vee, \wedge)$ is a complemented ~~lattice~~ distributive lattice, then De Morgan's Laws

(36)

$$(a \vee b)' = a' \wedge b' \quad \text{and} \quad (a \wedge b)' = a' \vee b'$$

hold for all $a, b \in L$.

proof: Observe that L is already bounded by definition of a complemented lattice. Also complements are unique because of distributivity.

To prove $(a \vee b)' = a' \wedge b'$, we simply prove that $(a' \wedge b')$ is complement of $(a \vee b)$ i.e., $(a \vee b) \vee (a' \wedge b') = 1$ and $(a \vee b) \wedge (a' \wedge b') = 0$.

We have,

$$\begin{aligned} (a \vee b) \vee (a' \wedge b') &= [(a \vee b) \vee a'] \wedge [(a \vee b) \vee b'] \quad \text{by distributivity} \\ &= [(a \vee a') \vee b] \wedge [a \vee (b \vee b')] \\ &= [1 \vee b] \wedge [a \vee 1] \\ &= 1 \wedge 1 = 1 \end{aligned}$$

$$\begin{aligned} \text{and } (a \vee b) \wedge (a' \wedge b') &= [(a \wedge (a' \wedge b'))] \vee [b \wedge (a' \wedge b')] \\ &= [(a \wedge a') \wedge b'] \vee [a' \wedge (b \wedge b')] \\ &= [0 \wedge b'] \vee [a' \wedge 0] \\ &= 0 \vee 0 = 0. \end{aligned}$$

That $(a' \wedge b')$ behaves as the complement of $(a \vee b)$. By uniqueness of complements, it is the only complement of $a \vee b$.

$$(a \vee b)' = a' \wedge b'.$$

The other follows dually.

Ex: Prove that a lattice $(L, \leq)$ is modular if and only if $(a \wedge b) \vee (a \wedge c) = a \wedge (b \vee (a \wedge c))$ for all $a, b, c \in L$. (37)

→ Suppose $(L, \leq)$ is modular. Then

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &= (a \wedge c) \vee (a \wedge b) && \text{by commutativity} \\ &= (a \wedge c) \vee (b \wedge a) && \text{by commutativity} \\ &= ((a \wedge c) \vee b) \wedge a && \text{by modularity since } a \wedge c \leq b \\ &= a \wedge (b \vee (a \wedge c)) \end{aligned}$$

Conversely, suppose that $(a \wedge b) \vee (a \wedge c) = a \wedge (b \vee (a \wedge c))$ for all $a, b, c \in L$.
Let $a, b, c \in L$ be such that $a \leq c$. Then $a \wedge c = a$.

$$\text{Now, } (c \wedge b) \vee (c \wedge a) = c \wedge (b \vee (a \wedge c)).$$

$$\text{Hence, } (c \wedge b) \vee a = c \wedge (b \vee a)$$

$$\text{i.e., } a \vee (b \wedge c) = (a \vee b) \wedge c$$

Hence (proved).

Ex: Show that in a complemented, distributive lattice, the followings are equivalent.

$$\text{i) } a \leq b \quad \text{ii) } a \wedge b' = 0 \quad \text{iii) } a' \vee b = 1 \quad \text{iv) } b' \leq a'$$

→

$$\begin{aligned} \text{i) } a \leq b &\Rightarrow a \vee b = b \\ &\Rightarrow (a \vee b) \wedge b' = 0 && [\because b \wedge b' = 0] \\ &\Rightarrow (a \wedge b') \vee (b \wedge b') = 0 && \text{by distributivity} \\ &\Rightarrow a \wedge b' = 0 \end{aligned}$$

Hence (i) $\Rightarrow$ (ii)

$$\begin{aligned} \text{Again } a \wedge b' = 0 &\Rightarrow (a \wedge b')' = 1 \\ &\Rightarrow a' \vee (b')' = 1 \\ &\Rightarrow a' \vee b = 1 \end{aligned}$$

Hence (ii) $\Rightarrow$ (iii)

$$\begin{aligned} \text{Now } a' \vee b = 1 &\Rightarrow (a' \vee b') \wedge b' = b' && [\because 1 \wedge b' = b'] \\ &\Rightarrow (a' \wedge b') \vee (b \wedge b') = b' && \text{by distributivity} \\ &\Rightarrow a' \wedge b' = b' && [\because b \wedge b' = 0] \\ &\Rightarrow b' \leq a' \end{aligned}$$

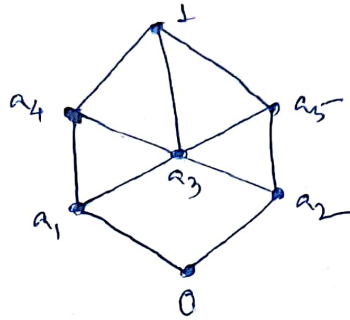
Hence (iii) $\Rightarrow$ (iv)

now, $b' \leq a' \Rightarrow a' \wedge b' = b'$
 $\Rightarrow (a' \wedge b')' = b$ by De Morgan's law
 $\Rightarrow a \vee b = b$
 $\Rightarrow a \leq b$

Here (iv) $\Rightarrow$ (i)

Thus, (i), (ii), (iii) and (iv) are equivalent.

Consider the lattice L in the figure given below



a) Which of the following are sublattices of L ?

$$L_1 = \{0, a_1, a_2, 1\}, L_2 = \{0, a_1, a_5, 1\}, L_3 = \{a_1, a_3, a_4, 1\}$$

b) Find complements, if exist, for elements a_1, a_2 and a_3 .

c) Is L distributive?

d) Is L a complemented lattice?

→

a) A subset L' is a sublattice if it is closed under $\wedge$ and $\vee$

L_1 is not a sublattice since $a_1 \vee a_2 = a_3$ which does not belong to L_1 . L_2 is a sublattice since $a_1 \vee a_5 = 1$ and $a_1 \wedge a_5 = 0 \in L_2$

L_3 is a sublattice since $a_3 \vee a_4 = 1$ and $a_3 \wedge a_4 = a_1$, $a_1 \vee a_3 = 1$ and $a_1 \wedge a_3 = a_4$, $a_1 \vee a_4 = 1$, $a_1 \wedge a_4 = a_3$.

b) We have $a_1 \wedge a_5 = 0$ and $a_1 \vee a_5 = 1$, so a_1 and a_5 are complements.

Also $a_2 \wedge a_4 = 0$ and $a_2 \vee a_4 = 1$, so a_2 and a_4 are complements.

But a_3 has no complement.

c) L is not distributive since $K = \{0, a_1, a_4, a_5, 1\}$ is a sublattice which forms pentagon.

d) L is not complemented since a_3 has no complement.